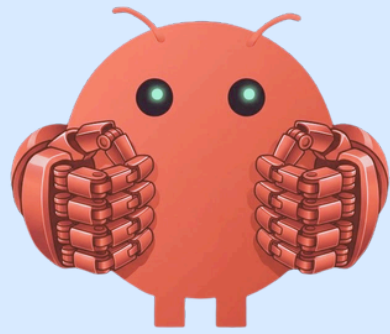




build the future of digital payments

FIN-O-HACK

fintech hackathon

Presented By:

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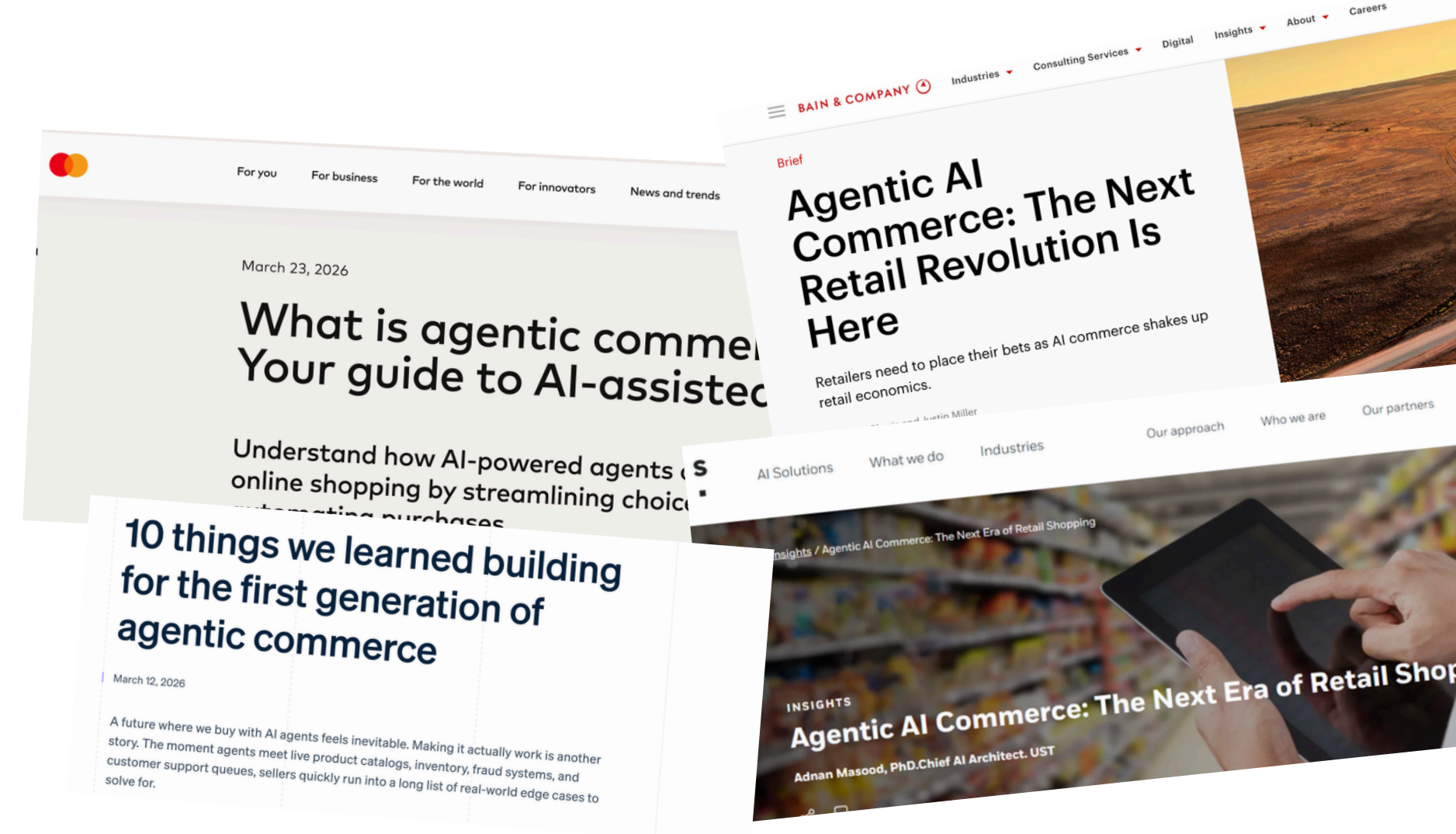
PROBLEM CONTEXT & BACKGROUND

What situation or observation led to this problem?

India's **63 million** small businesses run on **paper ledgers and manual GST filings**, completely outside the formal financial system. Paytm processes hundreds of **millions of transactions daily** yet none of that data flows back to help the merchants generating it. The same UPI transaction that could build a **shopkeeper's credit profile, reconcile their taxes, and predict their cash flow** does none of those things today. The infrastructure exists; the intelligence layer is missing.

WHY THE SYSTEM STALLS

- 01** Paytm captures rich merchant data like **POS sales and UPI transactions**, yet SMBs still **rely on CAs and face loan rejections** due to “no financial history.” At the same time, **78% file GST manually** with errors, and **over Rs. 20 trillion is stuck** in delayed B2B payments. The gap is not data, but the lack of integration.
- 02** Users drop off in **multi-step payment flows**, while SMBs remain invisible to AI-driven buyers. There is **no unified way** for users to **express intent** and get it executed securely, or for merchants to be discoverable to automated systems.
- 03** **CIBIL-based credit excludes** cash-driven SMBs. Without a transaction-based credit identity, **63 million businesses** remain locked out of formal lending and depend on informal sources. The necessary data already exists within Paytm but is not used for credit assessment.



ROOT CAUSES OF INEFFICIENCY



The Trust Deficit

Agentic AI lacks security primitives to operate on **real money safely**. No way to delegate payments without handing over **full credentials**. No audit trail — users can't verify what the AI did or why. **Malicious merchants** can manipulate AI agents via prompt injection.



The Fragmentation Maze

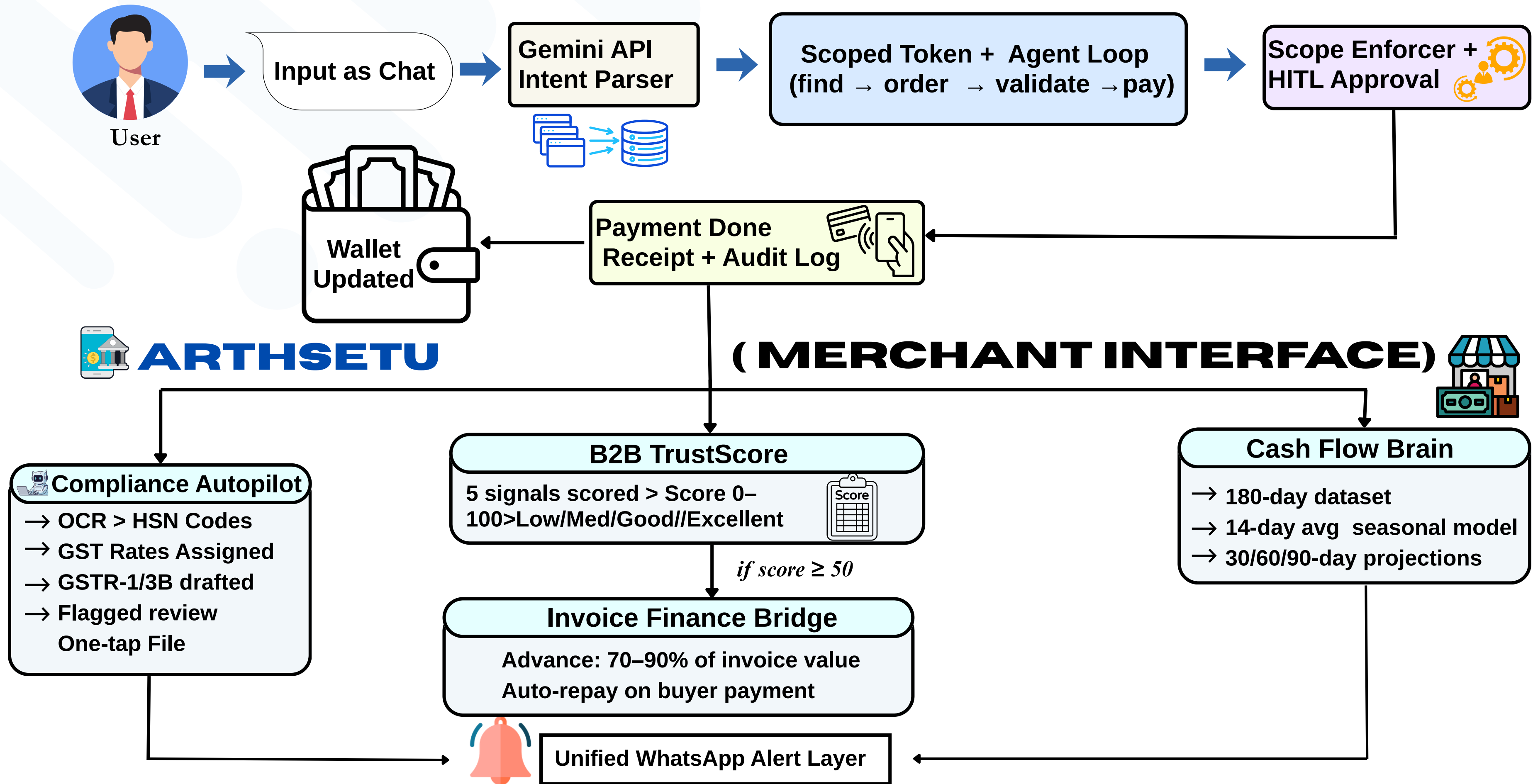
Users navigate **4+ siloed flows** for everyday tasks and abandon mid-journey. **No unified intent layer** can express a goal and have it executed. Consumer experience and merchant finances are **completely disconnected**. SMBs still run on WhatsApp, paper ledgers, and spreadsheets.



The Informal Finance Wall

63M SMBs invisible to formal credit despite daily Paytm transactions. **78% file GST manually** error-prone, CA dependent, unscalable. **₹20T stuck in late B2B** payments annually with no early recourse. Rich POS and UPI data goes entirely unused for credit assessment.

PAYBOT AGENTIC COMMERCE (USER INTERFACE)





ACTION STARTS HERE

Compliance Gst

1

Automates GST compliance from invoices and voice inputs using OCR, speech processing, and intelligent classification.

- OCR ingestion: Mistral AI (image/PDF → data)
- Voice interface: Whisper (STT) & Edge TTS
- Computes CGST / SGST / IGST
- Generates:Json required to be uploaded in the GST Portal
- Flags low-confidence cases
- Sends alerts: deadlines, liability, reminders

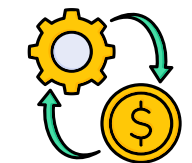


Cash Flow Brain

2

Predicts merchant revenue from historical data to enable proactive decisions like inventory planning and risk detection.

- Merchant wise time series
- 14-day rolling average + seasonal patterns (weekend, month-end)
- Forecasts: 30 / 60 / 90 days
- Outputs: Health (Good / Warning / Critical)
- Alerts for reorder & cashflow risk



B2B TrustScore Engine

3

Builds a real-time credit score for merchants from financial and compliance data.

- Score: 0–100 per merchant
- Based on 5 signals:payment behavior, consistency, revenue trend, GST compliance & dispute rate
- Updates after transactions & GST events
- Outputs: score + breakdown



Invoice Finance Bridge

4

Enables instant credit using TrustScore and transaction data to manage cash flow gaps.

- Detects overdue invoices (>15 days)
- If the TrustScore ≥ 50 then it becomes eligible
- Credit: 70–90% advance (tier-based limits)
- Instant disbursal + auto-repayment
- Integrated with transaction system



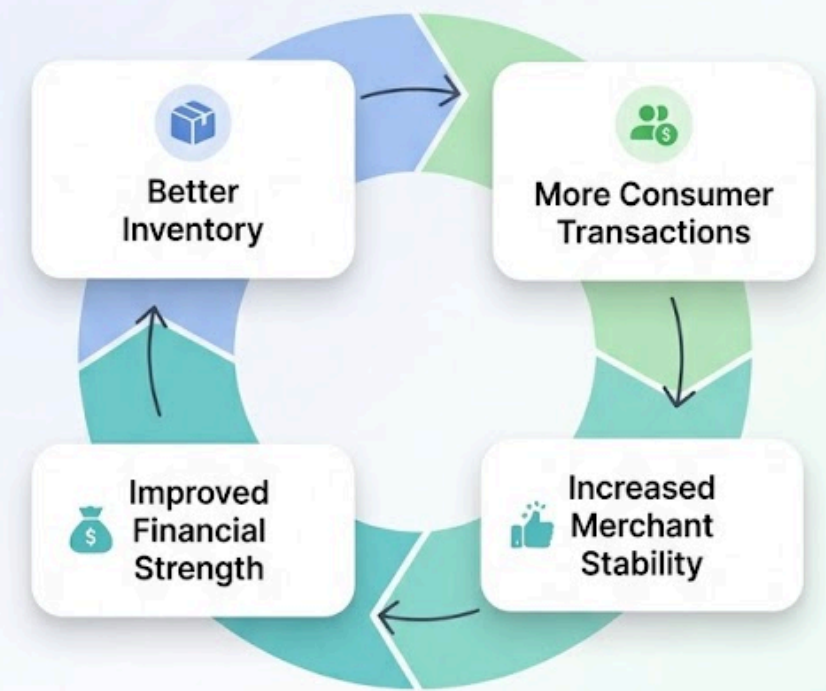
PayBot Commerce Agent

5

AI agent that converts natural language into secure transactions, optimizing across multiple merchants.

- Intent → structured actions (Gemini API)
- Scoped token:
 - budget · category · time
 - prepare order
 - Scope Enforcer + HITL (> ₹200)
- Agent loop:
 - search (Blinkit / Zepto / catalog) · Tokenized payment
 - compare prices → select cheapest execution
 - Outputs: receipt + audit log

The Result: The Compounding Flywheel



Projected Impact met with ArthSetu

	Before	After
Consumer Task Completion	60%	90%+
GST Compliance	30%	80%+
Credit Access	Baseline	5x Increase
Invoice Finance Turnaround	20 Days	< 4 Hours

ArthSetu creates a self-reinforcing growth loop for MSMEs

PAYBOT AGENTIC COMMERCE PIPELINE

1 VERIFIABLE INTENT : CRYPTOGRAPHIC DELEGATION TOKENS

Scoped intent credential minted before any agent action reaches payment rails, enforced at infrastructure level not model level.

- Bounded by amount, merchant category, and TTL
- Agent holds a revocable proof of delegated intent never user credentials
- Token revocation instantly strips all payment capability regardless of execution state

2 A2A PROTOCOL : DUAL-AGENT TRUST ISOLATION

Two agents, two independent trust scopes reasoning and execution are fully decoupled.

- Orchestrator Agent parses intent, sequences tool calls, zero payment authority
- Payment Agent sole access to payment rails, treats every Orchestrator instruction as untrusted, re-validates against intent token independently
- A compromised Orchestrator cannot escalate privileges into the Payment Agent

3 SERVER-SIDE SCOPE ENFORCEMENT + AUDIT TRAIL

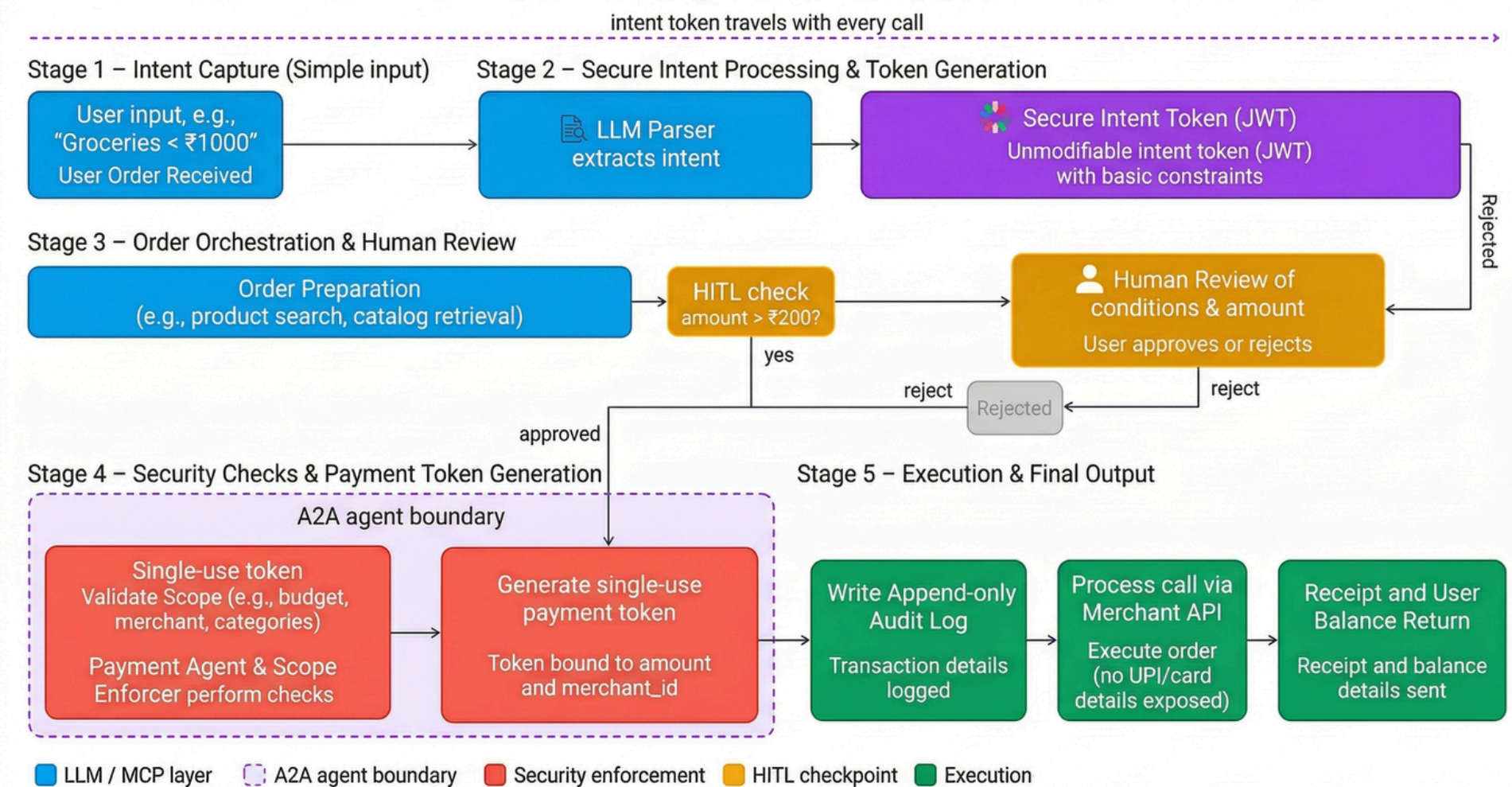
Payment validation lives entirely outside the agent cannot be bypassed, overridden, or reasoned.

- Backend independently re-validates every action against the original intent token before UPI rails are touched
- Every action written to an append-only audit log with timestamp and credential ID
- Continuous anomaly detection on audit stream triggers automatic agent freeze
- HITL gate holds execution on high-value transactions until the user approval is received

4 MCP TOOL BOUNDARIES : ZERO RAW EXECUTION SURFACE

Anthropic's Model Context Protocol enforces strict typed interfaces between the LLM and all backend services.

- Every agent capability is a formally defined tool with validated input/output schemas
- Only whitelisted MCP-registered tools are reachable no general execution access
- Merchant data processed in isolated sandboxes, fully separated from agent context
- Prompt injection is prevented no raw surface exists for malicious instructions to land



PAYBOT TECHNICAL FLOW

ARTHSETU: MERCHANT INTELLIGENCE LAYER

1 THE MULTI-AGENT COMPLIANCE SYSTEM

A coordinated multi-agent system that automates end-to-end GST filing without human intervention.

- **Orchestrator Agent** splits tasks and coordinates specialist agents in parallel
- **Invoice Agent** extracts structured data via OCR with confidence-based flagging
- **HSN Agent** classifies items and detects fraud patterns
- **Reconciliation Agent** matches invoices with UPI and GSTR-3B data
- **Draft Agent** prepares GST returns; **Filing Agent** submits securely with idempotency

2 EVENT-DRIVEN DATA BACKBONE

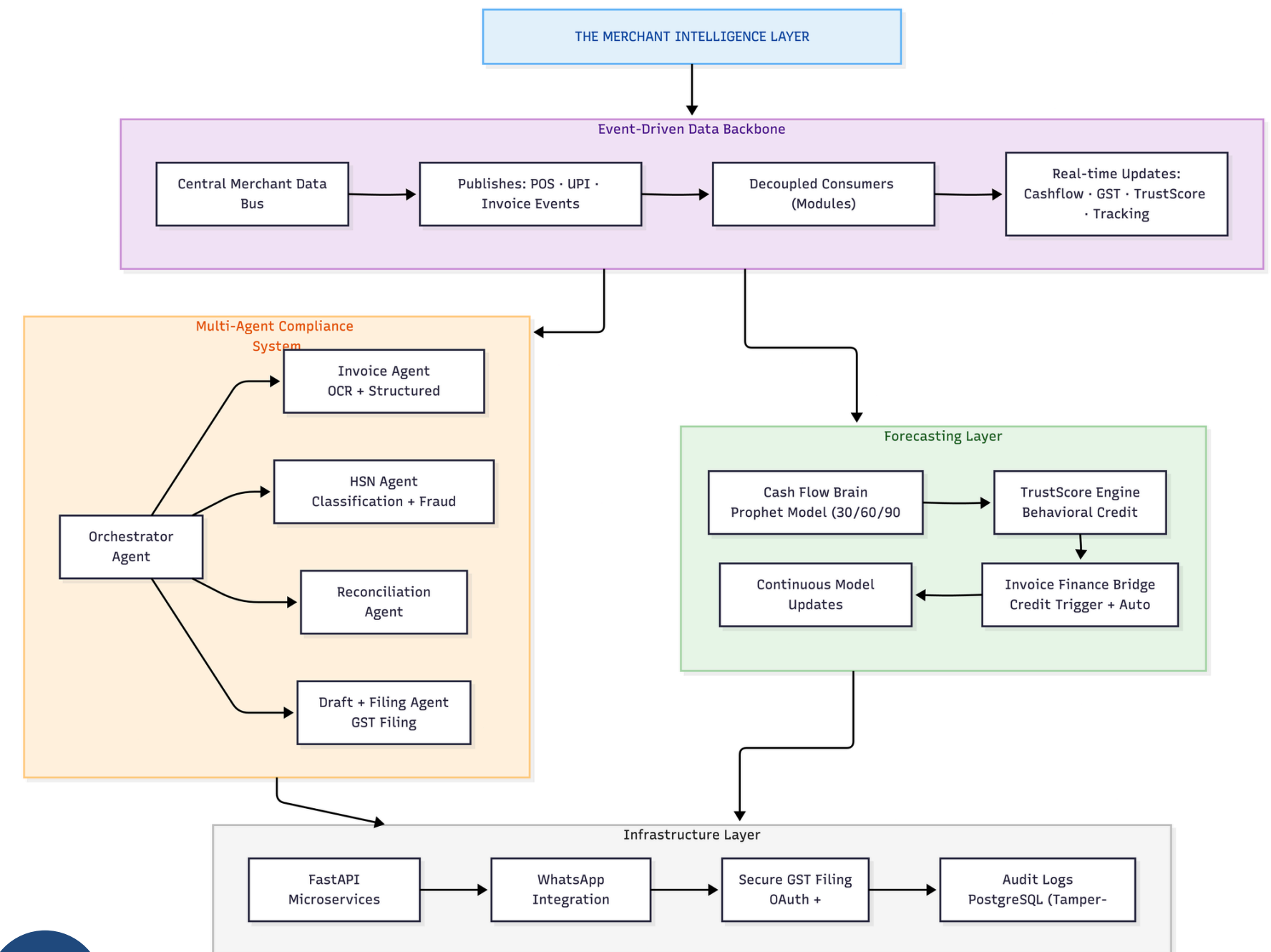
A real-time, decoupled data layer where every transaction powers all modules simultaneously.

- **Central Merchant Data Bus (Redis Streams)** publishes POS, UPI, and invoice
- **Modules** act as independent consumers with no direct coupling
- **Single transaction** updates cash flow, GST, TrustScore, invoice tracking, and AuraPay in real time

3 FORECASTING LAYER

AI-driven prediction and credit intelligence built on real transaction behaviour.

- **Cash Flow Brain** uses Prophet with seasonality to generate 30/60/90-day forecasts
- **TrustScore Engine** computes a multi-factor behavioural credit score normalised across peer groups
- **Invoice Finance Bridge** monitors invoices, triggers credit on delays, and auto-repays via UPI events
- **Models** update periodically using fresh transaction data for improved accuracy



4 INFRASTRUCTURE LAYER

A scalable, secure microservices architecture built for real-time financial operations.

- **FastAPI** microservices on Kubernetes with autoscaling based on stream load
- **WhatsApp** ingestion via Twilio with secure temporary storage and pluggable lending integration
- **GST** filing via OAuth with idempotent submissions and webhook confirmations
- **Tamper-proof** audit logs stored in PostgreSQL with strict access controls

THE PAYTM MOAT & BUSINESS IMPACT

THE MARKET OPPORTUNITY

\$150B+ SMB Fintech Market

India's SMB fintech market is projected to reach **\$150B+ by 2030** at 28% CAGR driven by digital payments, credit, and compliance automation.

₹25–30 Lakh Crore Credit Gap

India's SMB credit gap is among the largest in the world. Invoice delays alone lock up **₹20 trillion annually**, a direct target for the Invoice Finance Bridge.

GST Compliance Zero Automation

78% of SMBs file GST manually every quarter with frequent errors. The ₹1,200Cr+ compliance SaaS market has near-zero penetration today.

63M Unserved Small Businesses

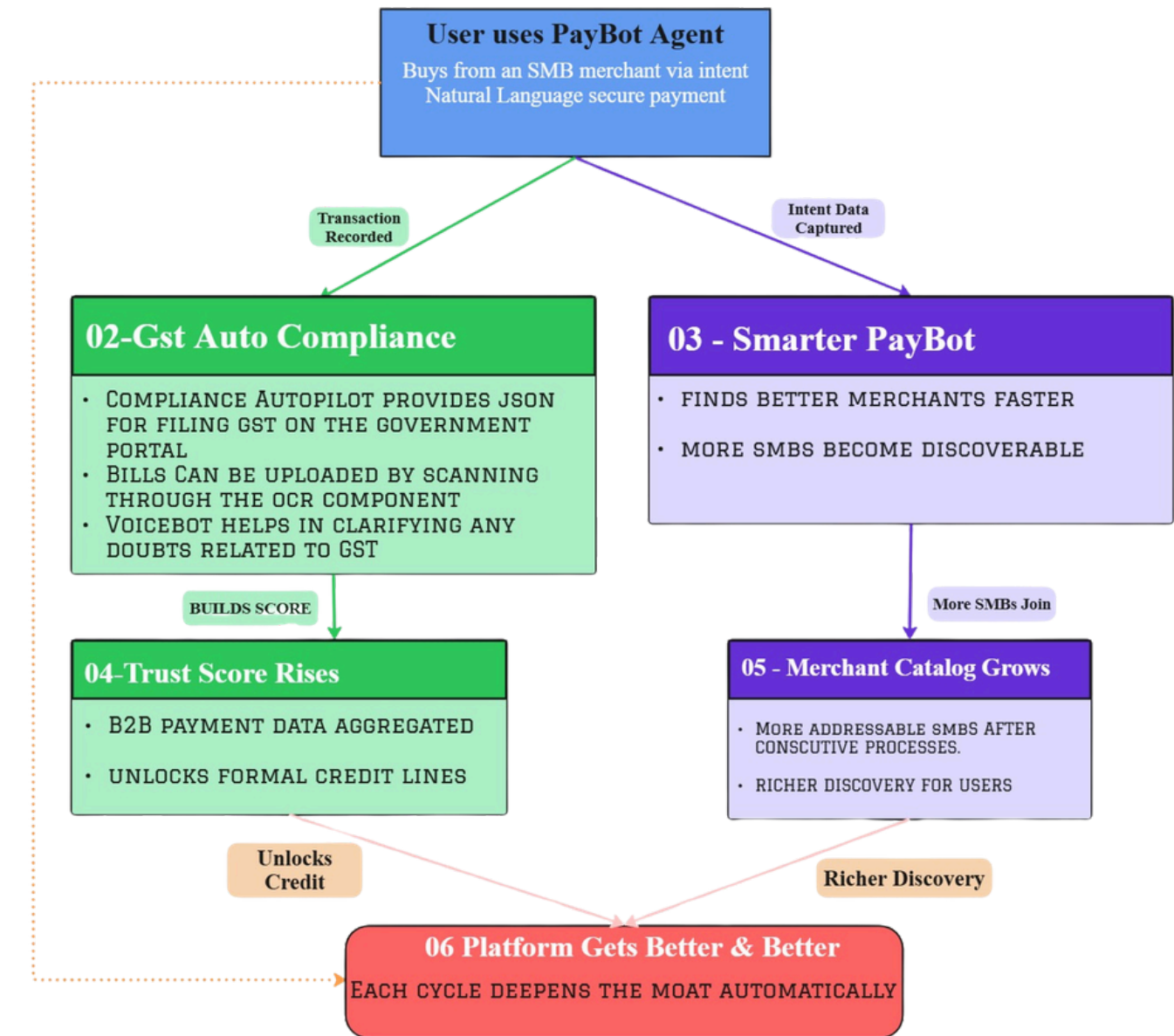
85% of India's 63M SMBs have never accessed formal credit. CIBIL-based systems exclude cash-driven merchants, MIL builds their financial identity from scratch.

\$2T+ Annual UPI Volume

UPI crossed \$2 trillion in annual transactions growing at 40% YoY. None of this data currently flows back to help the merchants generating it..

\$25B Agentic Payments by 2028

Global agentic AI payments is an emerging \$25B market. Aurapay positions Paytm as India's first mover, before any bank or aggregator captures the category.



“The 5 Structural Moats”

Bilateral Data

Both buyer & seller visible in every transaction

1

Merchant Network

Millions of pre-verified SMBs are instantly AI-addressable

2

UPI Rails

From intent to final payment in a single user session

3

Lending Arm

Quick access to credit: within 4 hours, no CIBIL check required

4

Two Sided KYC

Both users and merchants are pre-verified at the outset

5

No single competitor holds all 5 moats simultaneously
Compared to: Khatabook | Razorpay | PhonePe